

SOLIDWORKS API (.NET VBA C#) — Reference (273/337): ISketchManager (part 3 of 3)

SOLIDWORKS API Documentation (converted from CHM)

ISketchManager (part 3 of 3) (Type)

SOLIDWORKS API Help

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SetDynamicMirror Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : SetDynamicMirror Method (ISketchManager)

Enables or disables dynamic sketch mirroring, which is the automatic mirroring of newly created sketch entities about a selected centerline.

.NET Syntax

Visual Basic (Declaration)

```
Function SetDynamicMirror( _  
    ByVal DynamicMirror As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim DynamicMirror As System.Boolean  
Dim value As System.Boolean  
  
value = instance.SetDynamicMirror(DynamicMirror)
```

C#

```
System.bool SetDynamicMirror(  
    System.bool DynamicMirror  
)
```

C++/CLI

```
System.bool SetDynamicMirror(  
& System.bool DynamicMirror  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DynamicMirror

True to enable dynamic sketch mirroring, false to disable it

Return Value

True if setting this option to succeeds, false if fails

Visual Basic for Applications (VBA) Syntax

See [SketchManager::SetDynamicMirror](#).

Example

[Dynamically Mirror Sketch Entities \(VBA\)](#)

Remarks

If enabling dynamic sketch mirroring, then:

- a sketch must be in edit mode.
- a line segment or linear edge of a model must be selected before calling this method.

See Also

[ISketchManager Interface](#)

[ISketchManager Members](#)

[ISketchManager::GetDynamicMirror Method](#)

SetGridOptions Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : SetGridOptions Method (ISketchManager)

Sets the options for the grid.

Visual Basic Syntax

Visual Basic (Declaration)

```
Sub SetGridOptions( _
    ByVal Display As System.Boolean, _
    ByVal Spacing As System.Double, _
    ByVal Snap As System.Boolean, _
    ByVal DotStyle As System.Boolean, _
    ByVal NumMajor As System.Short, _
    ByVal NumMinor As System.Short, _
    ByVal AlignToEdge As System.Boolean, _
    ByVal AngleSnap As System.Boolean, _
    ByVal AngleUnit As System.Double, _
    ByVal MinorAuto As System.Boolean _
)
```

Visual Basic (Usage)



```
Dim instance As ISketchManager
Dim Display As System.Boolean
Dim Spacing As System.Double
Dim Snap As System.Boolean
Dim DotStyle As System.Boolean
Dim NumMajor As System.Short
Dim NumMinor As System.Short
Dim AlignToEdge As System.Boolean
Dim AngleSnap As System.Boolean
Dim AngleUnit As System.Double
Dim MinorAuto As System.Boolean

instance.SetGridOptions(Display, Spacing, Snap, DotStyle, NumMajor, NumMinor, AlignToEdge, AngleSnap, AngleUnit, MinorAuto)
```

C#

```
void SetGridOptions(
    System.bool Display,
    System.double Spacing,
    System.bool Snap,
    System.bool DotStyle,
    System.short NumMajor,
    System.short NumMinor,
    System.bool AlignToEdge,
    System.bool AngleSnap,
    System.double AngleUnit,
    System.bool MinorAuto
)
```

C++/CLI

```
void SetGridOptions(
    & System.bool Display,
    & System.double Spacing,
    & System.bool Snap,
    & System.bool DotStyle,
    & System.short NumMajor,
```

```

& System.short NumMinor,
& System.bool AlignToEdge,
& System.bool AngleSnap,
& System.double AngleUnit,
& System.bool MinorAuto
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Display
True to display the grid, false if not

Spacing
Snap distance

Snap
True to snap to grid, false to not

DotStyle
True for dotted grids, false if not

NumMajor
Number of minors in major

NumMinor
Number of snaps in minor

AlignToEdge
True if to align to an edge, false if not

AngleSnap
True to snap to angle, false to not

AngleUnit
Angle at which to snap

MinorAuto
True if the minor grids are to be set automatically, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::SetGridOptions](#).

Remarks

The AlignToEdge argument aligns the grid with the currently selected edge. If AlignToEdge is set to TRUE, then you must have selected an edge

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic](#).

SketchExtend Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : SketchExtend Method (ISketchManager)

Adds to the length of the selected sketch entity (i.e., line, centerline, or arc) to meet the nearest sketch entity.

.NET Syntax

Visual Basic (Declaration)

```

Function SketchExtend( _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double _
) As System.Boolean

```

Visual Basic (Usage)

```

Dim instance As ISketchManager
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim value As System.Boolean

```



```
value = instance.SketchExtend(X, Y, Z)
```

C#

```
System.bool SketchExtend(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.bool SketchExtend(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
Specify 0.0
Y
Specify 0.0
Z
Specify 0.0

Return Value

True if the sketch entity is extended to meet the nearest sketch entity, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::SketchExtend](#).

Example

[Extend Sketch Entity \(VBA\)](#)
[Extend Sketch Entity \(VB.NET\)](#)
[Extend Sketch Entity \(C#\)](#)

Remarks

Call this method multiple times to extend the selected line to meet successive sketch entities.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::SketchTrim Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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SketchOffset Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : SketchOffset Method (ISketchManager)

Obsolete. Superseded by [ISketchManager::SketchOffset2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SketchOffset( _  
    ByVal Offset As System.Double, _
```

```

ByVal BothDirections As System.Boolean, _
ByVal Chain As System.Boolean, _
ByVal CapEnds As System.Boolean, _
ByVal MakeConstruction As System.Boolean, _
ByVal AddDimensions As System.Boolean _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As ISketchManager
Dim Offset As System.Double
Dim BothDirections As System.Boolean
Dim Chain As System.Boolean
Dim CapEnds As System.Boolean
Dim MakeConstruction As System.Boolean
Dim AddDimensions As System.Boolean
Dim value As System.Boolean

value = instance.SketchOffset(Offset, BothDirections, Chain, CapEnds, MakeConstruction,
AddDimensions)

```

C#

```

System.bool SketchOffset(
    System.double Offset,
    System.bool BothDirections,
    System.bool Chain,
    System.bool CapEnds,
    System.bool MakeConstruction,
    System.bool AddDimensions
)

```

C++/CLI

```

System.bool SketchOffset(
    & System.double Offset,
    & System.bool BothDirections,
    & System.bool Chain,
    & System.bool CapEnds,
    & System.bool MakeConstruction,
    & System.bool AddDimensions
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Offset

Offset value; negative value offsets in opposite direction

BothDirections

True to offset the sketch entities in both directions, false to not

Chain

True to offset the chain of sketch entities, false if you want only the selected sketch entities offset (see **Remarks**)

CapEnds

True to cap the bidirectional offset with arcs, false to not

MakeConstruction

True to make the sketch entities construction geometry after offsetting, false to not

AddDimensions

True to include the offset distance in the sketch, false to not

Return Value

True if the sketch entities are offset, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::SketchOffset](#).

Remarks

Specifying true to the Chain argument offsets the selected entity and any other entities that belong to the same contour or chain (contiguous geometric entities like edges).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

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SketchOffset2 Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : SketchOffset2 Method (ISketchManager)

Offsets the selected sketch entities.

.NET Syntax

Visual Basic (Declaration)

```
Function SketchOffset2( _  
    ByVal Offset As System.Double, _  
    ByVal BothDirections As System.Boolean, _  
    ByVal Chain As System.Boolean, _  
    ByVal CapEnds As System.Integer, _  
    ByVal MakeConstruction As System.Integer, _  
    ByVal AddDimensions As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim Offset As System.Double  
Dim BothDirections As System.Boolean  
Dim Chain As System.Boolean  
Dim CapEnds As System.Integer  
Dim MakeConstruction As System.Integer  
Dim AddDimensions As System.Boolean  
Dim value As System.Boolean  
  
value = instance.SketchOffset2(Offset, BothDirections, Chain, CapEnds, MakeConstruction,  
AddDimensions)
```

C#

```
System.bool SketchOffset2(  
    System.double Offset,  
    System.bool BothDirections,  
    System.bool Chain,  
    System.int CapEnds,  
    System.int MakeConstruction,  
    System.bool AddDimensions  
)
```

C++/CLI

```
System.bool SketchOffset2(  
& System.double Offset,  
& System.bool BothDirections,  
& System.bool Chain,  
& System.int CapEnds,  
& System.int MakeConstruction,  
& System.bool AddDimensions  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Offset

Offset value; negative value offsets the sketch entities in the opposite direction

BothDirections

True to offset the sketch entities in both directions, false to offset the sketch entities in one direction

Chain

True to offset the chain of sketch entities, false to offset only the selected sketch entities (see **Remarks**)

CapEnds

Cap the ends as defined by [swSkOffsetCapEndType_e](#)

MakeConstruction

Convert original and offset sketch entities to construction sketch entities as defined by [swSkOffsetMakeConstructionType_e](#)

AddDimensions

True to include the offset distance in the sketch, false to not

Return Value

True if the selected sketch entities are offset, false if not

 Visual Basic for Applications (VBA) Syntax

See [SketchManager::SketchOffset2](#).

 Example

[Offset Sketch \(C#\)](#)

[Offset Sketch \(VB.NET\)](#)

[Offset Sketch \(VBA\)](#)

 Remarks

Specifying true for Chain offsets the selected sketch entities and any other sketch entities that belong to the same contour or chain (contiguous geometric entities like edges).

 See Also

[ISketchManager Interface](#)

[ISketchManager Members](#)

[IModelDoc2::SketchOffsetEntities2 Method \(\)](#)

[IModelDoc2::SketchOffsetEdges Method \(\)](#)

[IModelDocExtension::SketchOffsetOnSurface Method \(\)](#)

 Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchReplace Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : SketchReplace Method (ISketchManager)

Obsolete. Superseded by [ISketchManager::SketchReplace2](#).

 .NET Syntax

Visual Basic (Declaration)

```
Function SketchReplace( _  
    ByVal MakeConstruction As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim MakeConstruction As System.Boolean  
Dim value As System.Boolean  
  
value = instance.SketchReplace(MakeConstruction)
```

C#


```
System.bool SketchReplace(  
    System.bool MakeConstruction  
)
```

C++/CLI

```
System.bool SketchReplace(  
& System.bool MakeConstruction  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MakeConstruction

True to convert the replaced sketch entity to construction geometry, false to not

Return Value

True if the the sketch entity is successfully replaced, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::SketchReplace](#).

Remarks

Before calling this method, call [IModelDocExtension::SelectByID2](#) with Append set to true to select each of the following entities:

1. Sketch entity to be replaced.
2. Replacement sketch entity that does not reference downstream geometry or have references outside of the sketch.

After calling this method, call [ISketchManager::InsertSketch](#) to rebuild the model with the replacement sketch.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchReplace2 Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : SketchReplace2 Method (ISketchManager)

Replaces a sketch entity in a model with another sketch entity, preserving all references.

.NET Syntax

Visual Basic (Declaration)

```
Function SketchReplace2( _  
    ByVal MakeConstruction As System.Boolean, _  
    ByVal MakeContour As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim MakeConstruction As System.Boolean  
Dim MakeContour As System.Boolean  
Dim value As System.Boolean
```

```
value = instance.SketchReplace2(MakeConstruction, MakeContour)
```

C#

```
System.bool SketchReplace2(  
    System.bool MakeConstruction,  
    System.bool MakeContour  
)
```

C++/CLI

```
System.bool SketchReplace2(  
& System.bool MakeConstruction,  
& System.bool MakeContour  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MakeConstruction

True to convert the replaced sketch entity to construction geometry, false to delete it

MakeContour

True to make the replacement sketch entity a contour, false to not

Return Value

True if the the sketch entity is successfully replaced, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::SketchReplace2](#).

Example

[Replace Sketch Entity \(VBA\)](#)
[Replace Sketch Entity \(VB.NET\)](#)
[Replace Sketch Entity \(C#\)](#)

Remarks

Before calling this method, call [IModelDocExtension::SelectByID2](#) with Append set to true to select each of the following entities:

1. Sketch entity to be replaced.
2. Replacement sketch entity that does not reference downstream geometry or have references outside of the sketch.

After calling this method, call [ISketchManager::InsertSketch](#) to rebuild the model with the replacement sketch.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

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SketchTrim Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : SketchTrim Method (ISketchManager)

Trims the selected sketch entities.

.NET Syntax

Visual Basic (Declaration)

```
Function SketchTrim( _  
    ByVal Option As System.Integer, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Boolean
```

```

Dim instance As ISketchManager
Dim Option As System.Integer
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim value As System.Boolean

value = instance.SketchTrim(Option, X, Y, Z)

```

C#

```

System.bool SketchTrim(
    System.int Option,
    System.double X,
    System.double Y,
    System.double Z
)

```

C++/CLI

```

System.bool SketchTrim(
    & System.int Option,
    & System.double X,
    & System.double Y,
    & System.double Z
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Option

Sketch trim option as defined in [swSketchTrimChoice_e](#) (see **Remarks**)

X

x coordinate of pick location (see **Remarks**)

Y

y coordinate of pick location (see **Remarks**)

Z

z coordinate of pick location (for 3D sketch) (see **Remarks**)

Return Value

True if the sketch entities are trimmed, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::SketchTrim](#).

Example

[Trim Sketch Entities \(C#\)](#)

[Trim Sketch Entities \(VB.NET\)](#)

[Trim Sketch Entities \(VBA\)](#)

Remarks

Before calling this method, you must edit a sketch and select sketch segments to trim using [IModelDocExtension::SelectByID2](#).

Specify X, Y, and Z with:

- non-zero doubles only when Option is specified with swSketchTrimChoice_e.swSketchTrimEntityPoint.
- 0.0, 0.0, and 0.0 for all other Options.

Option swSketchTrimChoice_e...	Corresponds to Trim PropertyManager's...	Before calling this method, you must select...
swSketchTrimClosest	Trim to closest	One, and only one, sketch segment to be trimmed to its closest intersection with another sketch entity.
swSketchTrimCorner	Corner	Two, and only two, sketch segments to extend/trim to form a corner.
swSketchTrimEntities	Power trim	One or more (up to total number) sketch segments in the model, specifying their pick points. The sketch segments are trimmed to the sketch segments nearest the pick points. When you trim sketch segments using Power trim in the user interface, you drag your cursor across sketch segments, and the point at which the cursor crosses a sketch segment is where that sketch segment is trimmed. The API simulates this functionality by requiring you to select sketch segments by pick points, which are used as references when trimming the selected sketch segments.

Option swSketchTrimChoice_e...	Corresponds to Trim PropertyManager's...	Before calling this method, you must select...
swSketchTrimEntityPoint	Power trim	One and only one sketch segment; trims to a specific point on the sketch segment. Call this method, specifying non-zero X, Y, and Z, the point at which to trim the selected sketch segment. The specified point must lie on the sketch segment.
swSketchTrimInside	Trim away inside	At least three sketch segments: two to create the boundary, and one or more that intersect both sketch segments in the boundary. All selected sketch segments that intersect the boundary are trimmed inside the boundary segments.
swSketchTrimOutside	Trim away outside	At least three sketch segments: two to create the trim boundary, and one or more that intersect both sketch segments in the boundary. All selected sketch segments that intersect the boundary are trimmed outside the boundary segments.
swSketchTrimTwoEntities	Power trim	Two intersecting sketch segments; trims the first selected sketch segment to the second intersecting sketch segment. Use IModelDocExtension::SelectByID2 with Mark = 0 to select two sketch segments, specifying their pick points. The order in which the sketch segments are selected determines which one is trimmed. The first selected sketch segment will be trimmed on its pick point side to the second selected sketch segment.

See the **Trim Entities** topic in the SOLIDWORKS help for more information about trimming sketch entities.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::SketchExtend Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SketchUseEdge Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : SketchUseEdge Method (ISketchManager)

Obsolete. Superseded by [ISketchManager::SketchUseEdge2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SketchUseEdge( _
    ByVal Chain As System.Boolean _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchManager
Dim Chain As System.Boolean
Dim value As System.Boolean
```

```
value = instance.SketchUseEdge(Chain)
```

C#

```
System.bool SketchUseEdge(
    System.bool Chain
)
```

C++/CLI

```
System.bool SketchUseEdge(
    & System.bool Chain
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Chain

True if to convert the entire chain of entities, false to convert only the selected sketch entities (see **Remarks**)

Return Value

True if the sketch entities are created, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::SketchUseEdge](#).

Example

[Insert Sheet Metal Edge Flange \(VBA\)](#)
[Offset Selected Edges in Active Sketch \(VBA\)](#)

Remarks

Specifying true for the Chain argument converts the selected entity and any other entities that belong to the same contour or chain (contiguous, geometric entities like edges) on a sketch plane.

To display all sketch relations symbols, set [IModelDoc2::SetUserPreferenceToggle](#) swViewSketchRelations to true, which can adversely affect performance. To hide all sketch relation symbols, set IModelDoc2::SetUserPreferenceToggle swViewSketchRelations to false, which can improve performance and was the default setting prior to SOLIDWORKS 2005.

See [IModelDoc2::SketchOffsetEntities2](#) method for similar functionality.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchUseEdge2 Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : SketchUseEdge2 Method (ISketchManager)

Obsolete. Superseded by [ISketchManager::SketchUseEdge3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SketchUseEdge2( _  
    ByVal Chain As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim Chain As System.Boolean  
Dim value As System.Boolean  
  
value = instance.SketchUseEdge2(Chain)
```

C#

```
System.bool SketchUseEdge2(  
    System.bool Chain  
)
```

C++/CLI

```
System.bool SketchUseEdge2(  
    & System.bool Chain  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Chain

True to convert the entire chain of sketch entities, false to convert only the selected sketch entities (see **Remarks**)

Return Value

True if the sketch entities are created, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::SketchUseEdge2](#).

Example

[Convert Faces' Edges to Sketch Entities \(VB.NET\)](#)
[Convert Faces' Edges to Sketch Entities \(VBA\)](#)

Remarks

Specifying true for the Chain argument creates the selected sketch entity and any other sketch entities that belong to the same contour or chain (contiguous geometric sketch entities like sketch edges) on a sketch plane.

To display all sketch relations symbols, set [IModelDocExtension::SetUserPreferenceToggle](#) swViewSketchRelations to true, which can adversely affect performance. To hide all sketch relation symbols, set IModelDocExtension::SetUserPreferenceToggle swViewSketchRelations to false, which can improve performance and was the default setting prior to SOLIDWORKS 2005.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[IModelDoc2::SketchOffsetEntities2 Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

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SketchUseEdge3 Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : SketchUseEdge3 Method (ISketchManager)

Creates sketch entities on a sketch plane by projecting selected edges, loops, faces, curves, and external sketch contours.

.NET Syntax

Visual Basic (Declaration)

```
Function SketchUseEdge3( _  
    ByVal Chain As System.Boolean, _  
    ByVal InnerLoops As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim Chain As System.Boolean  
Dim InnerLoops As System.Boolean  
Dim value As System.Boolean
```

```
value = instance.SketchUseEdge3(Chain, InnerLoops)
```

C#

```
System.bool SketchUseEdge3(  
    System.bool Chain,  
    System.bool InnerLoops  
)
```

C++/CLI

```
System.bool SketchUseEdge3(  
& System.bool Chain,  
& System.bool InnerLoops  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Chain

True to convert an entire chain of sketch entities, false to convert only the selected sketch entities (see **Remarks**)

InnerLoops

True to convert the inner loops of the selected faces to sketch entities, false to not

Return Value

True if the sketch entities are created, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::SketchUseEdge3](#).

Example

[Convert Edges and Inner Loops of Face to Sketch Entities \(C#\)](#)
[Convert Edges and Inner Loops of Face to Sketch Entities \(VB.NET\)](#)
[Convert Edges and Inner Loops of Face to Sketch Entities \(VBA\)](#)

Remarks

Specifying true for the Chain argument creates the selected sketch entity and any other sketch entities that belong to the same sketch contour or chain (contiguous geometric entities like sketch edges) on the sketch plane.

To display all sketch relations symbols, set [IModelDocExtension::SetUserPreferenceToggle](#) swViewSketchRelations to true, which can adversely affect performance. To hide all sketch relation symbols, set IModelDocExtension::SetUserPreferenceToggle swViewSketchRelations to false, which can improve performance and was the default setting prior to SOLIDWORKS 2005.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[IModelDoc2::SketchOffsetEntities2 Method \(\)](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SplitClosedSegment Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : SplitClosedSegment Method (ISketchManager)

Splits the selected closed sketch segment into two sketch segments.

.NET Syntax

Visual Basic (Declaration)

```
Function SplitClosedSegment( _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal Z1 As System.Double, _  
    ByVal X2 As System.Double, _  
    ByVal Y2 As System.Double, _  
    ByVal Z2 As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim X1 As System.Double  
Dim Y1 As System.Double  
Dim Z1 As System.Double  
Dim X2 As System.Double  
Dim Y2 As System.Double
```

```
Dim Z2 As System.Double
Dim value As System.Object

value = instance.SplitClosedSegment(X1, Y1, Z1, X2, Y2, Z2)
```

C#

```
System.Object SplitClosedSegment(
    System.double X1,
    System.double Y1,
    System.double Z1,
    System.double X2,
    System.double Y2,
    System.double Z2
)
```

C++/CLI

```
System.Object^ SplitClosedSegment(
    & System.double X1,
    & System.double Y1,
    & System.double Z1,
    & System.double X2,
    & System.double Y2,
    & System.double Z2
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X1
X coordinate of first point
Y1
Y coordinate of first point
Z1
Z coordinate of first point
X2
X coordinate of second point
Y2
Y coordinate of second point
Z2
Z coordinate of second point

Return Value

Array of [sketch segments](#) of the now split formerly closed sketch segment

Visual Basic for Applications (VBA) Syntax

See [SketchManager::SplitClosedSegment](#).

Example

[Split Closed Sketch Segment \(VBA\)](#)
[Split Closed Sketch Segment \(VB.NET\)](#)
[Split Closed Sketch Segment \(C#\)](#)

Remarks

Before calling this method, be sure to disable snapping by either:

- De-selecting **System Options > Sketch > Relations/Snaps > Enable snapping**
- OR -
- Calling [ISldWorks::SetUserPreferenceToggle\(swUserPreferenceToggle_e.swSketchInference, false\)](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::SplitOpenSegment Method](#)

Availability

SplitOpenSegment Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : SplitOpenSegment Method (ISketchManager)

Splits the selected open sketch segment into two sketch segments.

.NET Syntax

Visual Basic (Declaration)

```
Function SplitOpenSegment( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Object  
  
value = instance.SplitOpenSegment(X, Y, Z)
```

C#

```
System.object SplitOpenSegment(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.Object^ SplitOpenSegment(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X coordinate of the point that splits the sketch segment in two
Y
Y coordinate of the point that splits the sketch segment in two
Z
Z coordinate of the point that splits the sketch segment in two

Return Value

Array of [sketch segments](#) of the now split formerly open sketch segment

Visual Basic for Applications (VBA) Syntax

See [SketchManager::SplitOpenSegment](#).

Example

[Split Open Sketch Segment \(VBA\)](#)
[Split Open Sketch Segment \(VB.NET\)](#)
[Split Open Sketch Segment \(C#\)](#)

Remarks

Before calling this method, be sure to disable snapping by either:

- De-selecting **System Options > Sketch > Relations/Snaps > Enable snapping**
- OR -
- Calling [ISldWorks::SetUserPreferenceToggle\(swUserPreferenceToggle_e.swSketchInference, false\)](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::SplitClosedSegment Method](#)

Availability

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